

Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

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Why do prediction tasks exposed to the same temporal shift suffer very different conformal coverage failures?

CENTRAL IDEA

Use SHAP concentration, the share of feature importance concentrated in the top feature, as a predeployment vulnerability diagnostic for gradient-boosted classifiers.

MAIN FINDING

In the COVID-19 supply-chain case study, eight tasks under the same temporal shift show coverage losses ranging from negligible to over 70 percentage points. Across 16 multiclass tasks in nine domains, concentration is associated with coverage loss (Spearman $\rho = 0.853$).

EVIDENCE AND SCOPE

An APS-score bound connects concentration to sensitivity under explicit assumptions. Additional experiments on nine external datasets show partial transfer and examine mechanism-specific responses, including changes to prediction sets, retraining and adaptive calibration.

LIMITS

The empirical association is not a universal coverage guarantee or proof of causality. Task samples are small and dependent. The diagnostic is most relevant to concentrated feature dependence in boosted trees; the exploratory 40% threshold needs prospective validation and does not transfer automatically to neural models.

TAKEAWAY

Before deployment, ask how concentrated the model's dependence is, as well as how much the inputs might shift.

한국어 핵심 요약

같은 외부 충격을 받아도 모델이 소수 특징에 얼마나 의존하는지에 따라 포함률 손실이 달라질 수 있습니다. SHAP 중요도의 집중도를 취약성 진단에 활용합니다. 관찰된 관계와 이론의 가정을 구분하며, 특정 모델 계열에서 얻은 진단을 보편적인 보장으로 확대하지 않습니다.

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